

1 Solution — GIAM 7.4

Problem 6

1.1 Problem

Prove the identity

$$\binom{n}{k} \binom{k}{r} = \binom{n}{r} \binom{n-r}{k-r}$$

combinatorially.

(Implicit: $0 \leq r \leq k \leq n$. If $r > k$ or $k > n$, both sides are 0 by the standard convention $\binom{a}{b} = 0$ for $b > a$ or $b < 0$, so the identity holds trivially.)

1.2 The Object Being Counted

Both sides will count the same set: the collection of *nested subset pairs*

$$\mathcal{P} := \{ (A, B) : B \subseteq A \subseteq \{1, 2, \dots, n\}, |A| = k, |B| = r \}.$$

An element of $\mathcal{P}$ is a triple of disjoint regions partitioning the universe $\{1, \dots, n\}$:

- B itself — the *inner* subset of size r .
- $A \setminus B$ — the part of A outside B , of size $k - r$.
- $\{1, \dots, n\} \setminus A$ — the part of the universe outside A , of size $n - k$.

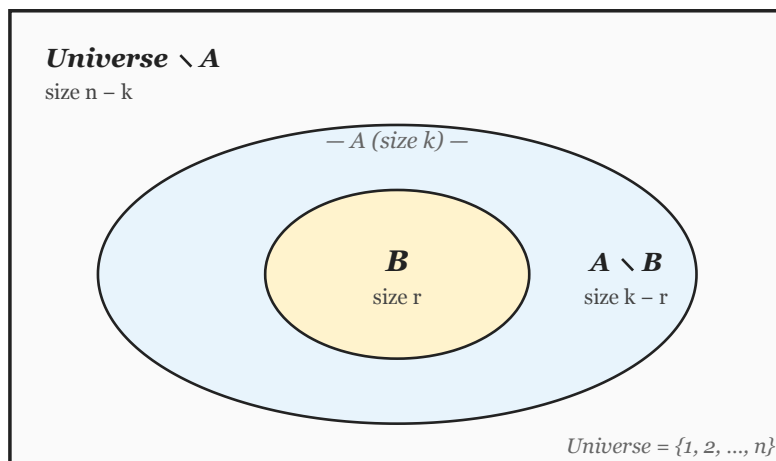


Figure 1.1: The combinatorial object: a pair (A, B) with $B \subseteq A \subseteq \{1, \dots, n\}$. Three regions partition the universe — B (size r), $A \setminus B$ (size $k - r$), and $Universe \setminus A$ (size $n - k$). Both sides of the identity count the number of such configurations.

The total sizes work out: $r + (k - r) + (n - k) = n$. ✓

The combinatorial proof shows $|\mathcal{P}|$ equals each side by *building* a nested pair in two different orders.

1.3 Counting Method 1 — Build A First, Then B

Construct $(A, B) \in \mathcal{P}$ in two stages:

- **Stage 1.** Choose the *outer* subset $A \subseteq \{1, \dots, n\}$ with $|A| = k$. Number of choices: $\binom{n}{k}$.
- **Stage 2.** Choose the *inner* subset $B \subseteq A$ with $|B| = r$. Number of choices: $\binom{k}{r}$.

Every nested pair (A, B) arises *exactly once* this way: A determines what's allowed in stage 2, and given A , the choice of B identifies the pair. By the multiplication rule,

$$|\mathcal{P}| = \binom{n}{k} \binom{k}{r}. \quad (*)$$

1.4 Counting Method 2 — Build B First, Then $A \setminus B$

Construct $(A, B) \in \mathcal{P}$ in a *different* two-stage order:

- **Stage 1'.** Choose the *inner* subset $B \subseteq \{1, \dots, n\}$ with $|B| = r$ directly from the full universe. Number of choices: $\binom{n}{r}$.
- **Stage 2'.** Complete the *outer* subset A by choosing the remaining $k - r$ elements of A from the elements *not* in B . There are $n - r$ such elements, and we choose $k - r$ of them. Number of choices: $\binom{n-r}{k-r}$. Then $A := B \cup (\text{the chosen } k - r \text{ elements})$.

Every nested pair $(A, B) \in \mathcal{P}$ arises *exactly once* this way: B is uniquely determined from the pair, and given B , the rest of A — namely $A \setminus B$ — is a $(k - r)$ -subset of $\{1, \dots, n\} \setminus B$. By the multiplication rule,

$$|\mathcal{P}| = \binom{n}{r} \binom{n-r}{k-r}. \quad (**)$$

1.5 Conclusion

Equations $(*)$ and $(**)$ count the same set $\mathcal{P}$, so

$$\boxed{\binom{n}{k} \binom{k}{r} = \binom{n}{r} \binom{n-r}{k-r}.}$$

■

1.6 Algebraic Cross-Check

Both sides equal the same product of factorials:

$$\binom{n}{k} \binom{k}{r} = \frac{n!}{k! (n-k)!} \cdot \frac{k!}{r! (k-r)!} = \frac{n!}{r! (k-r)! (n-k)!}.$$

$$\binom{n}{r} \binom{n-r}{k-r} = \frac{n!}{r! (n-r)!} \cdot \frac{(n-r)!}{(k-r)! (n-k)!} = \frac{n!}{r! (k-r)! (n-k)!}.$$

Both reduce to

$$\frac{n!}{r!(k-r)!(n-k)!} = \binom{n}{r, k-r, n-k},$$

the *trinomial coefficient* counting partitions of n elements into the three labeled groups of sizes r , $k-r$, and $n-k$. The combinatorial proof we just gave is *exactly* the statement that this trinomial counts $\mathcal{P}$.

1.7 Remark — Three-Way Symmetry

The trinomial form $\binom{n}{r, k-r, n-k}$ makes the symmetry of the identity transparent. The three labeled groups are interchangeable in the following sense: the same trinomial counts *any* partition of $\{1, \dots, n\}$ into three parts of the specified sizes. The identity in the problem corresponds to grouping the first two parts (B and $A \setminus B$) into A ; one could equally well group other pairs, yielding:

$$\binom{n}{r, k-r, n-k} = \binom{n}{k} \binom{k}{r} = \binom{n}{r} \binom{n-r}{k-r} = \binom{n}{n-k} \binom{k}{r}.$$

(The last form just uses $\binom{n}{n-k} = \binom{n}{k}$ — the standard complementary identity — and is the same as Method 1.) The two “essentially different” two-stage decompositions are the ones in the problem statement.

1.8 Remark — Special Cases

r	Identity simplifies to	Interpretation
0	$\binom{n}{k} \cdot 1 = \binom{n}{0} \binom{n}{k} = 1 \cdot \binom{n}{k}$	Both sides equal $\binom{n}{k}$; trivial.
1	$k \binom{n}{k} = n \binom{n-1}{k-1}$	<i>Absorption identity</i> — “every k -subset of $\{1, \dots, n\}$ has k distinguished elements, and there are n ways to pick a distinguished element together with $\binom{n-1}{k-1}$ ways to complete the rest.”
k	$\binom{n}{k} \cdot 1 = \binom{n}{k} \cdot 1$	Both sides equal $\binom{n}{k}$; identity collapses since $B = A$.

$k - 1$	$k \binom{n}{k} = \binom{n}{k-1} \binom{n-k+1}{1} = (n - k + 1) \binom{n}{k-1}$	A variant of the absorption identity.
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Table 1.1

The $r = 1$ case is especially common in proofs of derivative-like identities in combinatorics.

1.9 Remark — The Concrete Instance (Problem 5)

Problem 5 of this section is the *concrete* version of this identity with $n = 210$, $k = 24$, $r = 5$:

$$\underbrace{\binom{210}{24} \binom{24}{5}}_{\text{Outer-then-Inner}} = \underbrace{\binom{210}{5} \binom{205}{19}}_{\text{Inner-then-Outer}}.$$

Two formulas for *governance structures* — pairs (G, S) with $S \subseteq G \subseteq \text{Student Body}$ — built in two different orders. The general identity above is what *forces* the two formulas to agree.

1.10 Remark — On the Phrase “Combinatorially”

A “combinatorial proof” of an identity $\text{LHS} = \text{RHS}$ exhibits a *single* finite set whose size is counted by LHS *and* by RHS — typically by *two different orderings* of the same choice process. The advantage over an algebraic proof is conceptual: the formula’s *meaning* (what it counts) is laid bare, often with no factorials in sight.

The technique is sometimes called **double counting**, and is one of the most reliable methods for finding and proving combinatorial identities. The general schema:

1. Identify (or invent) a set $\mathcal{S}$ that the identity could count.
2. Find one way to build elements of $\mathcal{S}$ that yields LHS.
3. Find a *different* way to build elements of $\mathcal{S}$ that yields RHS.
4. Conclude $\text{LHS} = |\mathcal{S}| = \text{RHS}$.

For this problem, the set was $\mathcal{P}$ (nested pairs), and the two build orders were “outer then inner” and “inner then outer.”